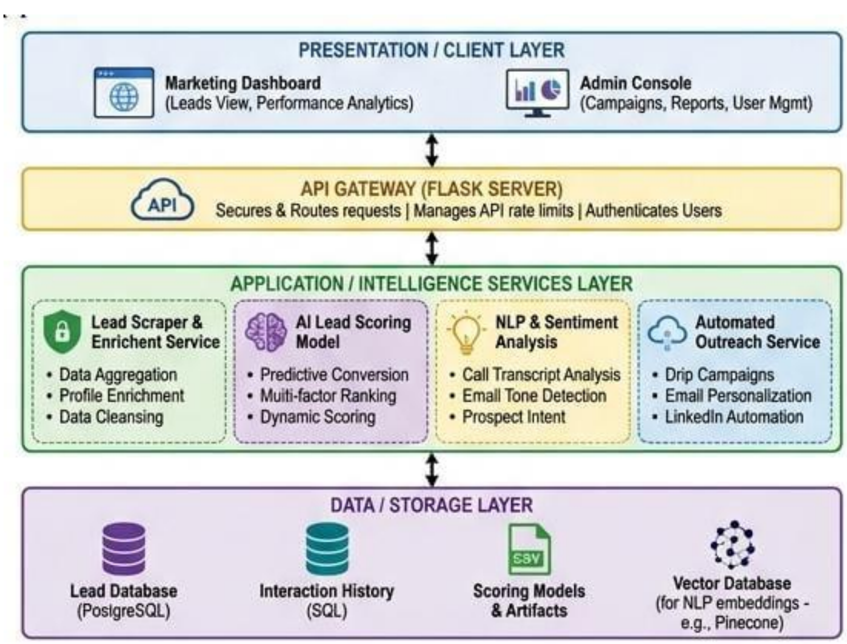
Solution Architecture

Date	28 June 2026
Team ID	xxxxxx
Project Name	AI-Powered Lead Generation System
Maximum Marks	5 Marks

Solution Architecture Diagram:

The **AI-Powered Lead Generation System** follows a layered architecture consisting of the **Presentation Layer, API/Service Layer, AI & Machine Learning Layer, and Data/Storage Layer**. The system allows sales and marketing users to enter customer or lead-related information through a web interface, where the data is securely processed by the backend. The application preprocesses the data, analyzes customer information using Machine Learning models, predicts lead quality, assigns lead scores, and generates intelligent recommendations for potential customers. All lead details, prediction results, and user activity logs are stored in the database for future analysis and improvement. This architecture ensures scalability, security, maintainability, efficient lead processing, and accurate AI-driven decision-making.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly web interface where users can enter customer details, manage leads, view lead scores, predictions, and recommendations.	HTML, CSS, JavaScript, Streamlit, Bootstrap
API Gateway	Receives requests from the frontend, validates input data, and routes requests to the appropriate backend	Flask (Python) / REST API
Authentication Service	Handles user registration, login, authentication, and role-based access for secure system usage.	Flask, Python, JWT/Session Authentication
Lead Data Processing Service	Uses trained Machine Learning models to analyze customer information, predict lead quality, assign lead scores, and identify high-potential prospects.	Python, Scikit-learn, Machine Learning Models
Recommendation Service	Generates intelligent recommendations and prioritizes leads based on prediction results and customer behavior analysis.	Python, AI/ML Algorithms
External Integration Service	Integrates with external customer databases, CRM systems, APIs, or other business tools for additional lead information.	REST API, External Data Sources
Data / Storage Layer	Stores customer details, lead history, prediction results, trained models, datasets, and system logs for future analysis.	SQLite, CSV Files, Database Systems, File System